LOW-LEVEL DESIGN (LLD) SPECIFICATIONS

HORIZON GROUP OF COMPANIES

Document Control & Reference Architecture Metadata

Project Type: Academic Case Study / Capstone Infrastructure Simulation

Document ID: HGC-HQ-LLD-2025-V1

Prepared By: Princess E.

Target Environment: Production Simulation (Cisco IOS Architecture)

Date of Issue: May 5, 2025

Classification: Technical Reference Architecture

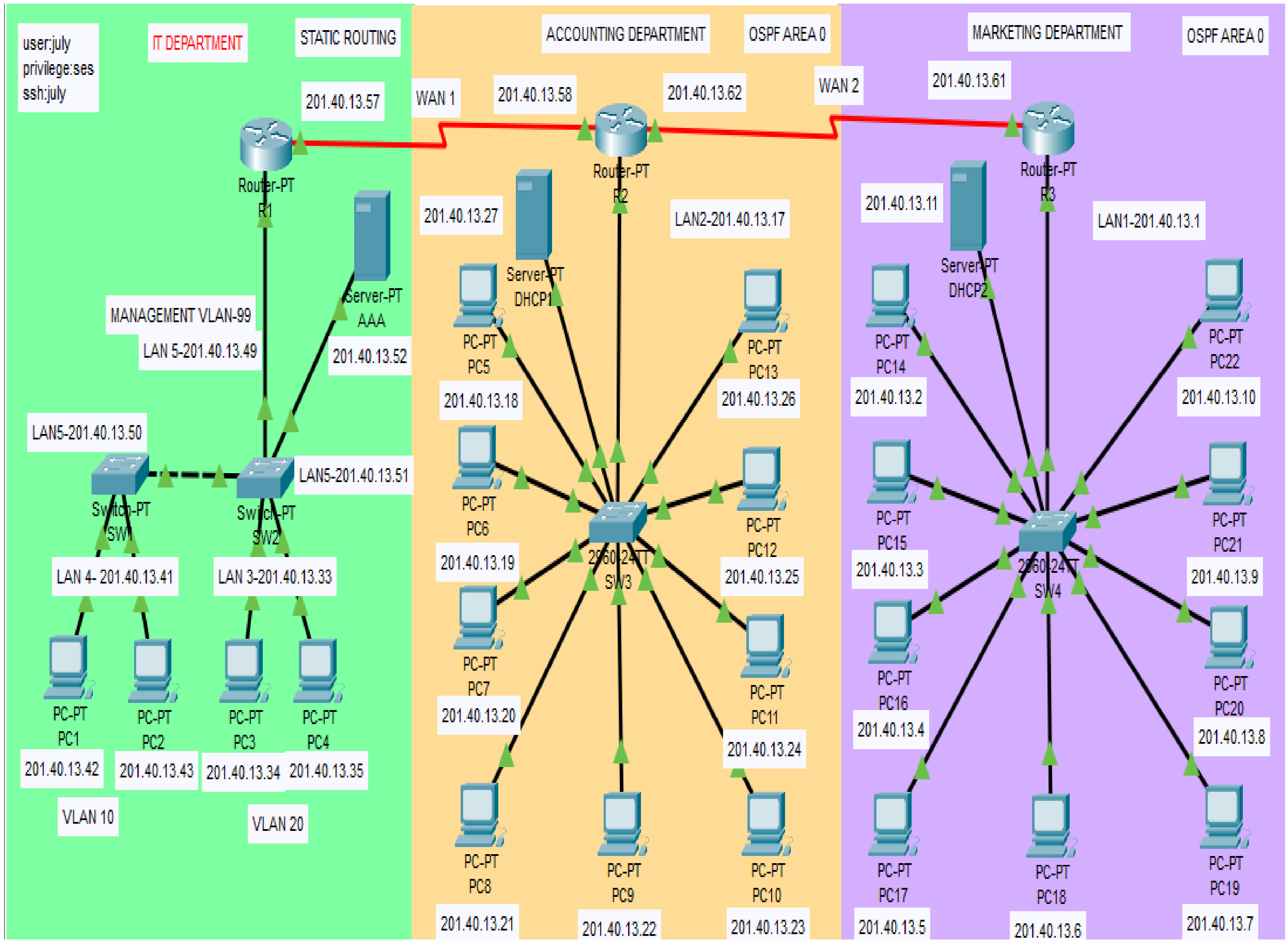
Document Revision History

v0.1 | April 5, 2025 | P. J. Escanan: Initial network architecture draft and VLSM calculations.

v0.2 | April 10, 2025 | P. J. Escanan: Integrated dynamic OSPF routing, DHCP pools, and TACACS+ AAA configurations.

v1.0 | April 20, 2025 | P. J. Escanan: Implemented VLANs onto Departments

TOPOLOGY



user:july
privilege:ses
ssh:july
IT DEPARTMENT
STATIC ROUTING
201.40.13.57
Router-PT
R1
MANAGEMENT VLAN-99
LAN 5-201.40.13.49
Server-PT
AAA
201.40.13.52
LAN5-201.40.13.50
LAN5-201.40.13.51
Switch-PT
SW1
Switch-PT
SW2
LAN 4- 201.40.13.41
LAN 3-201.40.13.33
PC-PT
PC1
201.40.13.42
PC-PT
PC2
201.40.13.43
PC-PT
PC3
201.40.13.34
PC-PT
PC4
201.40.13.35
VLAN 10
VLAN 20
WAN 1
ACCOUNTING DEPARTMENT
OSPF AREA 0
201.40.13.58
201.40.13.62
Router-PT
R2
201.40.13.27
Server-PT
DHCP1
LAN2-201.40.13.17
PC-PT
PC5
201.40.13.18
PC-PT
PC6
201.40.13.19
PC-PT
PC7
201.40.13.20
PC-PT
PC8
201.40.13.21
PC-PT
PC9
201.40.13.22
PC-PT
PC10
201.40.13.23
PC-PT
PC11
201.40.13.24
PC-PT
PC12
201.40.13.25
PC-PT
PC13
201.40.13.26
2960-24TT
SW3
WAN 2
MARKETING DEPARTMENT
OSPF AREA 0
201.40.13.61
Router-PT
R3
201.40.13.11
Server-PT
DHCP2
LAN1-201.40.13.1
PC-PT
PC14
201.40.13.2
PC-PT
PC15
201.40.13.3
PC-PT
PC16
201.40.13.4
PC-PT
PC17
201.40.13.5
PC-PT
PC18
201.40.13.6
PC-PT
PC19
201.40.13.7
PC-PT
PC20
201.40.13.8
PC-PT
PC21
201.40.13.9
PC-PT
PC22
201.40.13.10
2960-24TT
SW4

VLSM TABLE

VLSM FOR IP ADD: 201.40.13.0					
NETWORK USED	NETWORK ID	USABLE IP RANGE	BROADAST ID	SUBNET MASK	HOST AVAILABLE PER SUBNET
LAN 1	201.40.13.0	201.40.13.1 - 201.40.13.14	201.40.13.15	/28-255.255.255.240	16-2 = 14 hosts
LAN 2	201.40.13.16	201.40.13.17 - 201.40.13.30	201.40.13.31	/28-255.255.255.240	16-2 = 14 hosts
LAN 3	201.40.13.32	201.40.13.33 - 201.40.13.38	201.40.13.39	/29-255.255.255.248	8-2=6 hosts
LAN 4	201.40.13.40	201.40.13.41 - 201.40.13.46	201.40.13.47	/29-255.255.255.248	8-2=6 hosts
LAN 5	201.40.13.48	201.40.13.49 - 201.40.13.54	201.40.13.55	/29-255.255.255.248	8-2=6 hosts
WAN 1	201.40.13.56	201.40.13.57 - 201.40.13.58	201.40.13.59	/30-255.255.255.252	4-2=2 hosts
WAN 2	201.40.13.60	201.40.13.61 - 201.40.13.62	201.40.13.63	/30-255.255.255.252	4-2=2 hosts

Computations for LAN

Formula For Number of Subnets

2^n

n = number of host bits borrowed

Soln': 2^4 = 2*2*2*2 = 16 subnet

Formula For Number of Hosts Per Subnet

$2^h - 2$

h= Number of Remaining Host Bits

Given IP Address	201.40.13.0
Given IP Address Space	C
Number of Host Bits Available	8
Number of Host Bits Borrowed	4
Number of Host Bits Remaining (h)	8-4 = 4
Number of Possible Hosts	2^4 = 2*2*2*2 = 16 16-2 = 14
New Subnet Mask	255.255.255.240
New Prefix	/28

Calculate New Subnets by Using LSB (Least Significant Bit)

Subnet Mask(Dotted Decimal Format)	**Subnet Mask(Binary Format)**	**4th Octet Expanded**							
255.255.255.240	**11111111.11111111.11111111. 11110000**	**2^7**	**2^6**	**2^5**	**2^4**	**2^3**	**2^2**	**2^1**	**2^0**
		128	**64**	**32**	**16**	**8**	**4**	**2**	**1**
		1	**1**	**1**	**1 LSB**	**0**	**0**	**0**	**0**

Computations for LAN

Formula For Number of Subnets

2^n

n = number of host bits borrowed

Soln': 2^3 = 2*2*2 = 8 subnet

Formula For Number of Hosts Per Subnet

2^h - 2

h= Number of Remaining Host Bits

Given IP Address	201.40.13.32
Given IP Address Space	C
Number of Host Bits Available	4
Number of Host Bits Borrowed	1
Number of Host Bits Remaining (h)	4-1 = 3
Number of Possible Hosts	2^3 = 2*2*2 = 8 8-2 =6
New Subnet Mask	255.255.255.248
New Prefix	/29

Calculate New Subnets by Using LSB (Least Significant Bit)

Subnet Mask(Dotted Decimal Format)	**Subnet Mask(Binary Format)**	**4th Octet Expanded**							
255.255.255.240	**11111111.11111111.11111111. 11110000**	**2^7**	**2^6**	**2^5**	**2^4**	**2^3**	**2^2**	**2^1**	**2^0**
		128	**64**	**32**	**16**	**8**	**4**	**2**	**1**
		1	**1**	**1**	**1 LSB**	**1 NEW LSB**	**0**	**0**	**0**

Computations for WAN

Formula For Number of Subnets

2^n

n = number of host bits borrowed

Soln': 2^2 = 2*2= 4 subnet

Formula For Number of Hosts Per Subnet

2^h - 2

h= Number of Remaining Host Bits

Given IP Address	201.40.13.56
Given IP Address Space	C
Number of Host Bits Available	4
Number of Host Bits Borrowed	2
Number of Host Bits Remaining (h)	4-2 = 2
Number of Possible Hosts	2^2 = 2*2= 4 4-2 = 2
New Subnet Mask	255.255.255.252
New Prefix	/30

Calculate New Subnets by Using LSB (Least Significant Bit)

Subnet Mask(Dotted Decimal Format)	Subnet Mask(Binary Format)	4th Octet Expanded							
255.255.255.240	11111111.11111111.11111111.11110000	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
		128	64	32	16	8	4	2	1
		1	1	1	1 LSB	1 LSB	1 NEW LSB	0	0

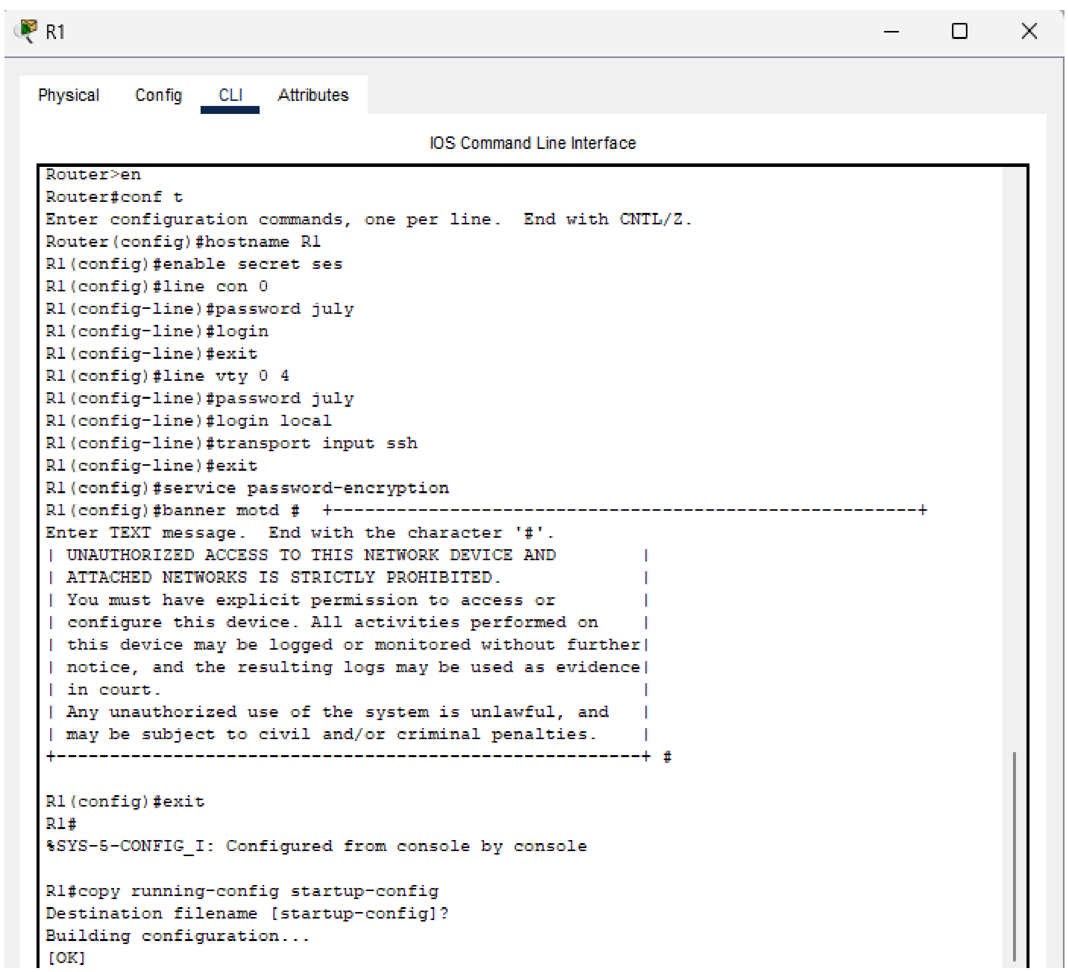
IP ADDRESS TABLE

AREA	DEVICE	PORT	IP ADD	SUBNET MASK	GATEWAY ADDRESS	WILDCARD MASK
LAN 1	PC14	NIC	201.40.13.2	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC15	NIC	201.40.13.3	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC16	NIC	201.40.13.4	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC17	NIC	201.40.13.5	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC18	NIC	201.40.13.6	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC19	NIC	201.40.13.7	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC20	NIC	201.40.13.8	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC21	NIC	201.40.13.9	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	PC22	NIC	201.40.13.10	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	DHCP2	NIC	201.40.13.11	/28-255.255.255.240	201.40.13.1	0.0.0.15
LAN 1	SW4	F0/1	201.40.13.1	/28-255.255.255.240	N/A	N/A
WAN 2	R3	F0/0	201.40.13.1	/28-255.255.255.240	N/A	0.0.0.15
		S2/0	201.40.13.61	/30-255.255.255.252	N/A	0.0.0.3
LAN 2	PC5	NIC	201.40.13.18	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC6	NIC	201.40.13.19	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC7	NIC	201.40.13.20	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC8	NIC	201.40.13.21	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC9	NIC	201.40.13.22	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC10	NIC	201.40.13.23	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC11	NIC	201.40.13.24	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC12	NIC	201.40.13.25	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	PC13	NIC	201.40.13.26	/28-255.255.255.240	201.40.13.17	0.0.0.15
LAN 2	DHCP1	NIC	201.40.13.27	/28-255.255.255.240	201.40.13.17	0.0.0.15
WAN 2	R2	S3/0	201.40.13.62	/30-255.255.255.252	N/A	0.0.0.3
WAN 1		S2/0	201.40.13.58	/30-255.255.255.252	N/A	0.0.0.3
LAN 2		F0/0	201.40.13.17	/28-255.255.255.240	N/A	0.0.0.15
LAN 2	SW3	F0/1	201.40.13.17	/28-255.255.255.240	N/A	0.0.0.15
LAN 4	PC1	NIC	201.40.13.42	/29-255.255.255.248	201.40.13.41	0.0.0.7
LAN 4	PC2	NIC	201.40.13.43	/29-255.255.255.248	201.40.13.41	0.0.0.7
LAN 3	PC3	NIC	201.40.13.34	/29-255.255.255.248	201.40.13.33	0.0.0.7
LAN 3	PC4	NIC	201.40.13.35	/29-255.255.255.248	201.40.13.33	0.0.0.7
LAN 5	SW1	F0/1	201.40.13.50	/29-255.255.255.248	201.40.13.49	0.0.0.7
LAN 5	SW2	F0/1	201.40.13.51	/29-255.255.255.248	201.40.13.49	0.0.0.7
		F1/1	201.40.13.49	/29-255.255.255.248	N/A	0.0.0.7
LAN 5	AAA	NIC	201.40.13.52	/29-255.255.255.248	201.40.13.49	0.0.0.7
WAN 1	R1	S2/0	201.40.13.57	/30-255.255.255.252	N/A	0.0.0.3
		F1/0	201.40.13.49	/29-255.255.255.248	N/A	0.0.0.7

VLAN TABLE

VLAN TABLE					
DEVICE	VLAN	INTERFACE	IP ADDRESS	SUBNET MASK	DEFAULT GATEWAY
R3	VLAN 10	F1/0.10	201.40.13.41	/29-255.255.255.248	N/A
	VLAN 20	F1/0.20	201.40.13.33		N/A
	VLAN 99	F1/0.99	201.40.13.49		N/A
SW1	VLAN 10	F2/1	201.40.13.42		201.40.13.41
		F1/1	201.40.13.43		201.40.13.41
	VLAN 99	VLAN 99	201.40.13.50		201.40.13.49
SW2	VLAN 20	F3/1	201.40.13.35		201.40.13.33
		F2/1	201.40.13.34		201.40.13.33
	VLAN 99	VLAN 99	201.40.13.51		201.40.13.49
PC1	VLAN 10	NIC	201.40.13.42		201.40.13.41
PC2			201.40.13.43		201.40.13.41
PC3	VLAN 20		201.40.13.34		201.40.13.33
PC4			201.40.13.35		201.40.13.33
AAA SERVER	VLAN 99		201.40.13.52		201.40.13.49

R1 BASIC CONFIGURATIONS




```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable secret ses
R1(config)#line con 0
R1(config-line)#password july
R1(config-line)#login
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password july
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#service password-encryption
R1(config)#banner motd #  +--------------------------------------------------------+
Enter TEXT message.  End with the character '#'.
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND        |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.             |
| You must have explicit permission to access or        |
| configure this device. All activities performed on    |
| this device may be logged or monitored without further|
| notice, and the resulting logs may be used as evidence|
| in court.                                             |
| Any unauthorized use of the system is unlawful, and   |
| may be subject to civil and/or criminal penalties.    |
+--------------------------------------------------------+ #

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

R2 BASIC CONFIGURATIONS

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>EN
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#enable secret ses
R2(config)#line con 0
R2(config-line)#password july
R2(config-line)#login
R2(config-line)#exit
R2(config)#line vty 0 4
R2(config-line)#password july
R2(config-line)#login local
R2(config-line)#transport input ssh
R2(config-line)#exit
R2(config)#banner motd #+-------------------------------------------------------+
Enter TEXT message.  End with the character '#'.
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND        |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.             |
| You must have explicit permission to access or        |
| configure this device. All activities performed on    |
| this device may be logged or monitored without further|
| notice, and the resulting logs may be used as evidence|
| in court.                                             |
| Any unauthorized use of the system is unlawful, and   |
| may be subject to civil and/or criminal penalties.    |
+-------------------------------------------------------+#

R2(config)#service password-encryption
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
```

R3 BASIC CONFIGURATIONS

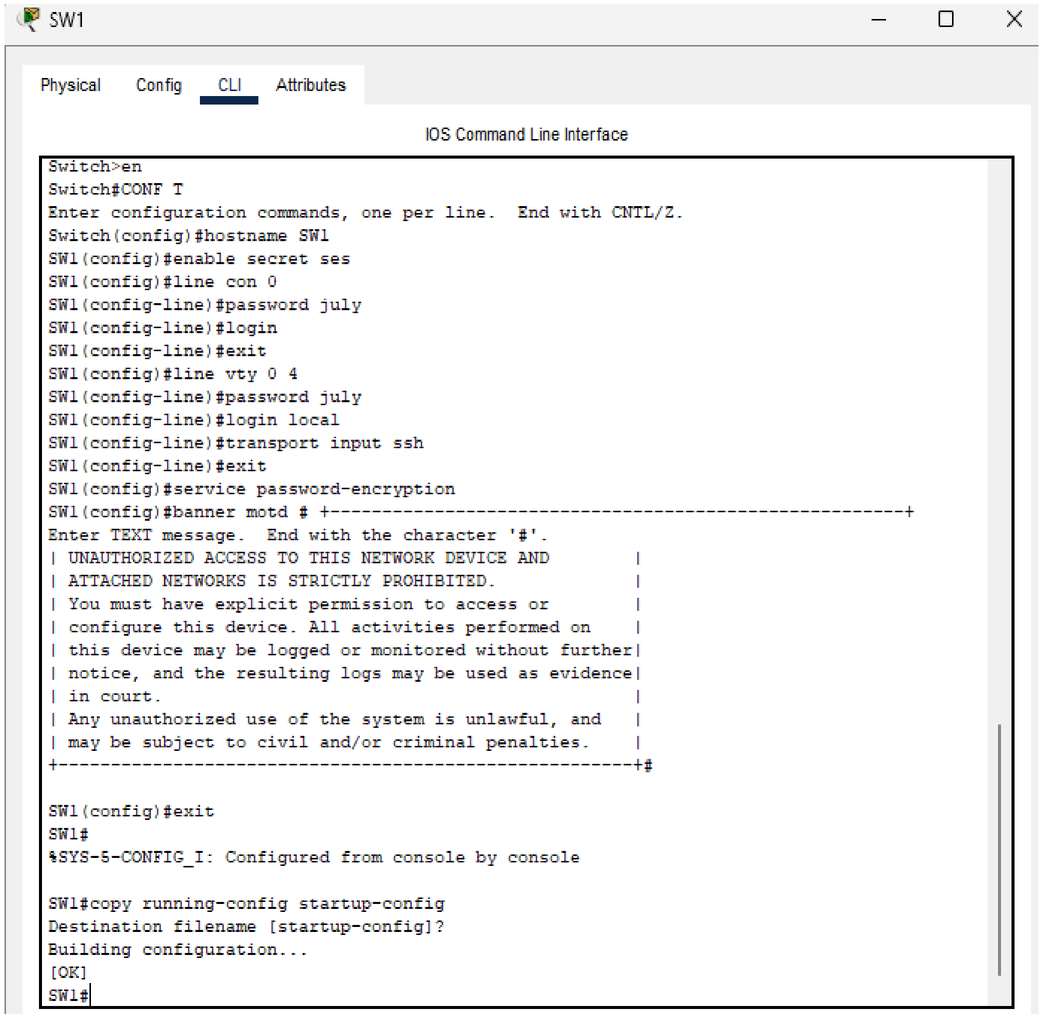
R3

Physical Config CLI Attributes

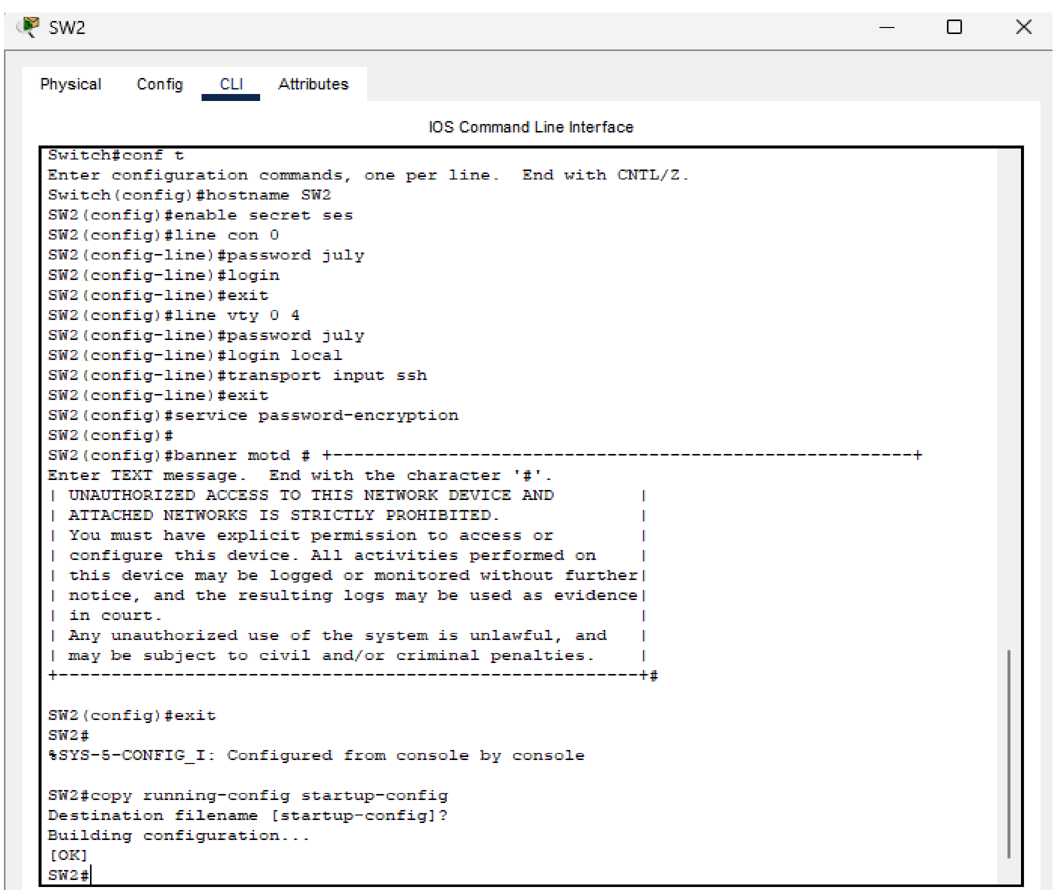
IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#enable secret ses
R3(config)#line con 0
R3(config-line)#password july
R3(config-line)#login
R3(config-line)#exit
R3(config)#line vty 0 4
R3(config-line)#password july
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exit
R3(config)#banner motd #+-------------------------------------------------------+
Enter TEXT message.  End with the character '#'.
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND        |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.             |
| You must have explicit permission to access or        |
| configure this device. All activities performed on    |
| this device may be logged or monitored without further|
| notice, and the resulting logs may be used as evidence|
| in court.                                             |
| Any unauthorized use of the system is unlawful, and   |
| may be subject to civil and/or criminal penalties.    |
+-------------------------------------------------------+#

R3(config)#service password-encryption
R3(config)#
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#
```

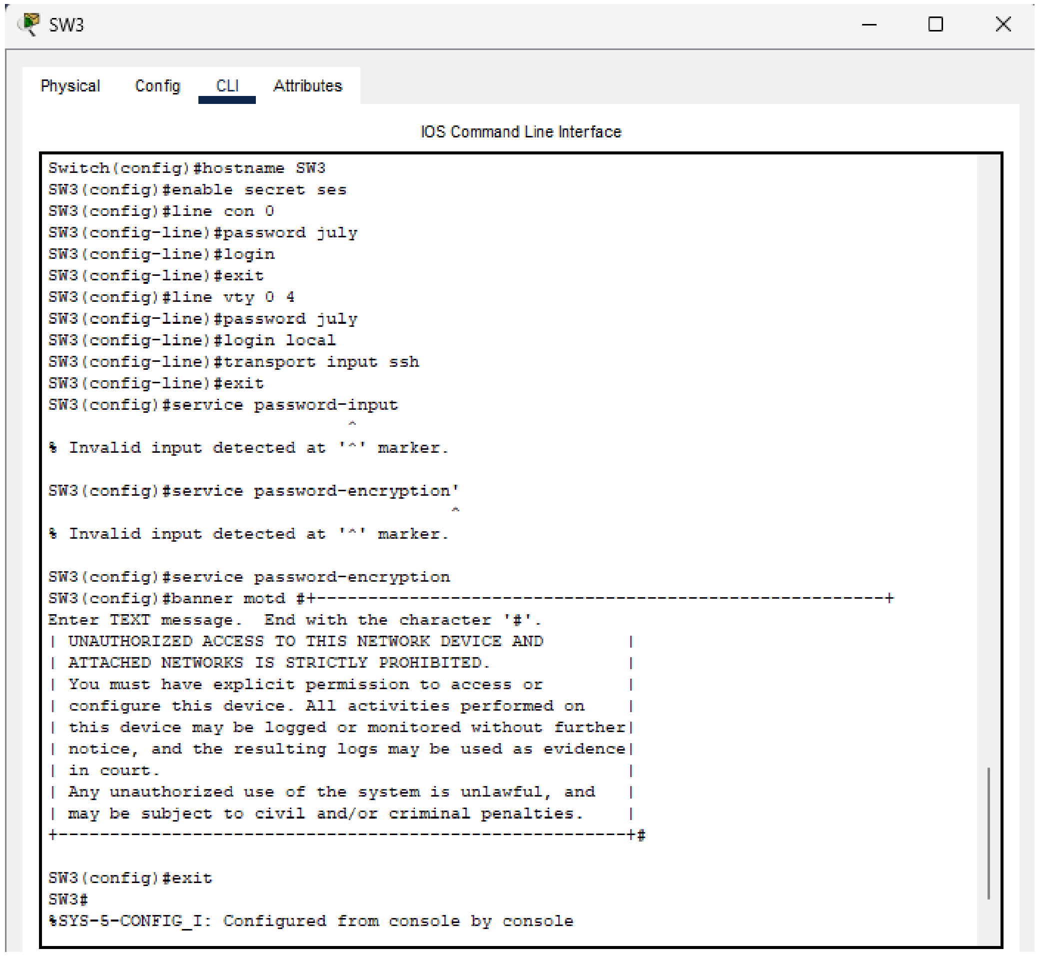
SW1 BASIC CONFIGURATIONS



SW2 BASIC CONFIGURATIONS

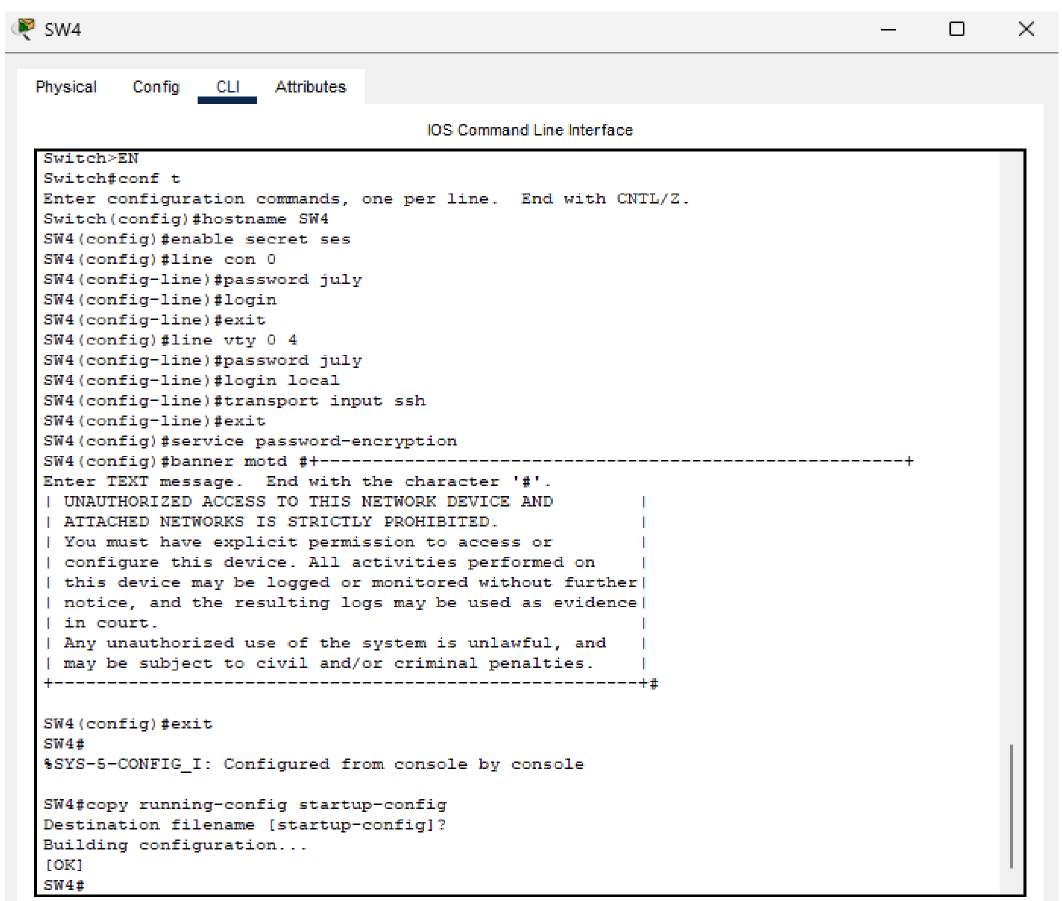


SW3 BASIC CONFIGURATIONS



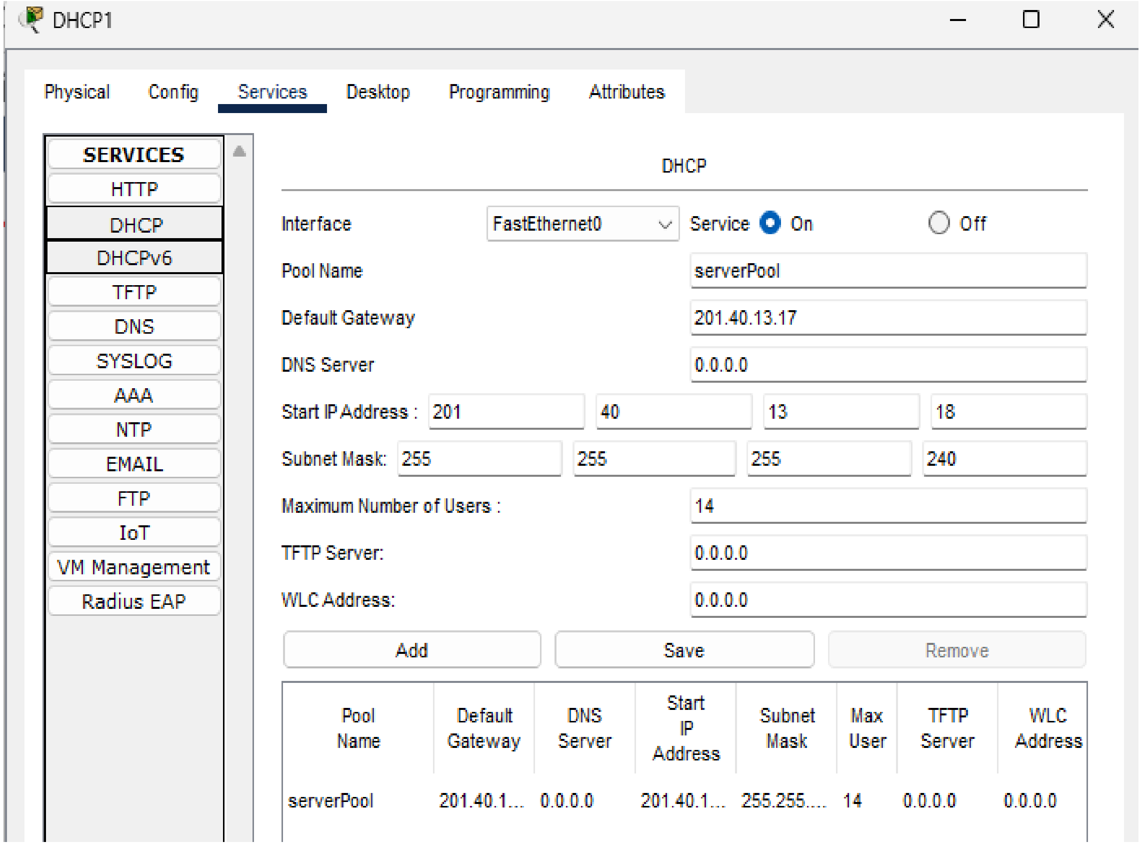

```
Switch(config)#hostname SW3
SW3(config)#enable secret ses
SW3(config)#line con 0
SW3(config-line)#password july
SW3(config-line)#login
SW3(config-line)#exit
SW3(config)#line vty 0 4
SW3(config-line)#password july
SW3(config-line)#login local
SW3(config-line)#transport input ssh
SW3(config-line)#exit
SW3(config)#service password-input
                                  ^
% Invalid input detected at '^' marker.

SW3(config)#service password-encryption'
                                       ^
% Invalid input detected at '^' marker.

SW3(config)#service password-encryption
SW3(config)#banner motd #+-------------------------------------------------------+
Enter TEXT message.  End with the character '#'.
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND        |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.             |
| You must have explicit permission to access or        |
| configure this device. All activities performed on    |
| this device may be logged or monitored without further|
| notice, and the resulting logs may be used as evidence|
| in court.                                             |
| Any unauthorized use of the system is unlawful, and   |
| may be subject to civil and/or criminal penalties.    |
+-------------------------------------------------------+#

SW3(config)#exit
SW3#
%SYS-5-CONFIG_I: Configured from console by console
```

SW4 BASIC CONFIGURATIONS

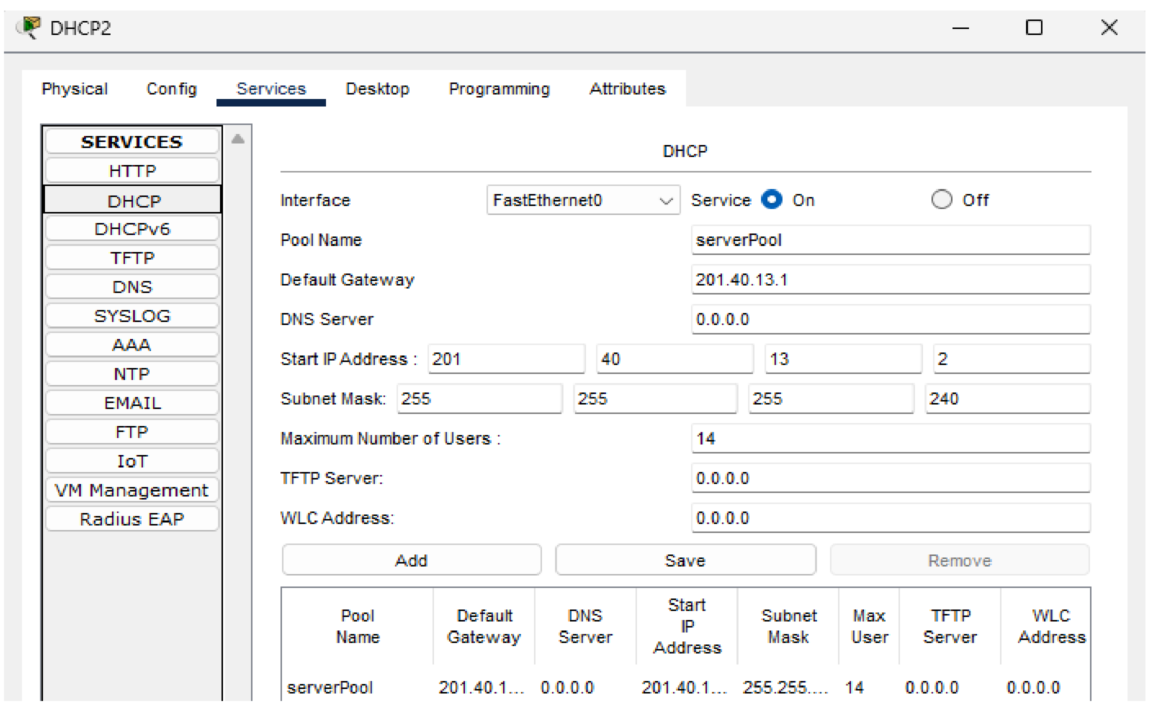



```
Switch>EN
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW4
SW4(config)#enable secret ses
SW4(config)#line con 0
SW4(config-line)#password july
SW4(config-line)#login
SW4(config-line)#exit
SW4(config)#line vty 0 4
SW4(config-line)#password july
SW4(config-line)#login local
SW4(config-line)#transport input ssh
SW4(config-line)#exit
SW4(config)#service password-encryption
SW4(config)#banner motd #+-------------------------------------------------------+
Enter TEXT message.  End with the character '#'.
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND        |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.             |
| You must have explicit permission to access or        |
| configure this device. All activities performed on    |
| this device may be logged or monitored without further|
| notice, and the resulting logs may be used as evidence|
| in court.                                             |
| Any unauthorized use of the system is unlawful, and   |
| may be subject to civil and/or criminal penalties.    |
+-------------------------------------------------------+#

SW4(config)#exit
SW4#
%SYS-5-CONFIG_I: Configured from console by console

SW4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW4#
```

DHCP1 SERVER CONFIGURATIONS



DHCP2 SERVER CONFIGURATIONS



R3 OSPF CONFIGURATION

```
R3(config-if)#exit
R3(config)#router ospf 10
R3(config-router)#network 201.40.13.0 0.0.0.15 area 0
R3(config-router)# network 201.40.13.60 0.0.0.3 area 0
R3(config-router)#passive-interface FastEthernet0/0
R3(config-router)#
00:08:53: %OSPF-5-ADJCHG: Process 10, Nbr 201.40.13.62 on Serial2/0 from LOADING to FULL,
Loading Done
```

R2 OSPF CONFIGURATION

```
R2(config)#router ospf 10
R2(config-router)#network 201.40.13.16 0.0.0.15 area 0
R2(config-router)# network 201.40.13.60 0.0.0.3 area 0
R2(config-router)#
00:08:53: %OSPF-5-ADJCHG: Process 10, Nbr 201.40.13.1 on Serial3/0 from LOADING to FULL,
Loading Done
passive-interface FastEthernet0/0
R2(config-router)#passive-interface FastEthernet0/0
```

R1 VLAN AND INTER-VLAN CONFIGURATIONS

R1

```
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f1/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip add 201.40.13.41 255.255.248
                                             ^
% Invalid input detected at '^' marker.

R1(config-subif)#ip add 201.40.13.41 255.255.255.248
R1(config-subif)#exit
R1(config)#int f1/0.20
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip add 201.40.13.33 255.255.255.248
R1(config-subif)#exit
R1(config)#int f1/0.99
R1(config-subif)#encapsulation dot1q 99
R1(config-subif)#ip add 201.40.13.49 255.255.255.248
R1(config-subif)#
```

Copy Paste

SW2 VLAN AND INTER-VLAN CONFIGURATIONS

```
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#vlan 10
SW2(config-vlan)#exit
SW2(config)#vlan 20
SW2(config-vlan)#exit
SW2(config)#vlan 99
SW2(config-vlan)#exit
SW2(config)#int vlan 99
SW2(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

SW2(config-if)#ip add 201.40.13.51 255.255.255.248
SW2(config-if)#no shut
SW2(config-if)#exit
SW2(config)#ip default-gateway 201.40.13.49
SW2(config)#int range f3/1,f2/1
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 20
SW2(config-if-range)#exit
SW2(config)#int range f1/1,f0/1
SW2(config-if-range)#switchport mode trunk
```

SW2

```
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int f4/1
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 99
SW2(config-if)#
```

SW1 VLAN AND INTER-VLAN CONFIGURATIONS

```
SW1(config)#vlan 10
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#exit
SW1(config)#vlan 99
SW1(config-vlan)#exit
SW1(config)#int vlan 99
SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

SW1(config-if)#ip add 201.40.13.50 255.255.255.248
```

```
SW1(config)#
SW1(config)#ip default-gateway 201.40.13.49
SW1(config)#int range f2/1,f1/1
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 10
SW1(config-if-range)#no shut
SW1(config-if-range)#exit
SW1(config)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/1
```

```
SW1(config)#int f0/1
SW1(config-if)#switchport mode trunk
SW1(config-if)#
```

R1 STATIC ROUTE CONFIGURATION

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 201.40.13.16 255.255.255.240 s2/0
R1(config)#R1(config)#
R1(config)#interface FastEthernet1/0
R1(config-if)#
R1(config-if)#exit
R1(config)#interface Serial2/0
R1(config-if)#exit
R1(config)#ip route 201.40.13.0 255.255.255.240 s2/0
R1(config)#ip route 201.40.13.62 255.255.255.252 s2/0
%Inconsistent address and mask
R1(config)#no ip route 201.40.13.62 255.255.255.252 s2/0
%Inconsistent address and mask
R1(config)#no ip route 201.40.13.60 255.255.255.252 s2/0
%No matching route to delete
R1(config)#ip route 201.40.13.60 255.255.255.252 s2/0
R1(config)#
```

R2 STATIC ROUTE CONFIGURATION

```
R2>en
Password:
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 201.40.13.49 255.255.255.252 s2/0
%Inconsistent address and mask
R2(config)#ip route 201.40.13.48 255.255.255.252 s2/0
R2(config)#no ip route 201.40.13.48 255.255.255.252 s2/0
R2(config)#no ip route 201.40.13.48 255.255.255.248 s2/0
%No matching route to delete
R2(config)#ip route 201.40.13.48 255.255.255.248 s2/0
R2(config)#ip route 201.40.13.32 255.255.255.248 s2/0
R2(config)#ip route 201.40.13.40 255.255.255.248 s2/0
R2(config)#exit
```

R3 STATIC ROUTE CONFIGURATIONS

```
R3>en
Password:
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 201.40.13.48 255.255.255.248 s2/0
R3(config)#ip route 201.40.13.32 255.255.255.248 s2/0
R3(config)#ip route 201.40.13.40 255.255.255.248 s2/0
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

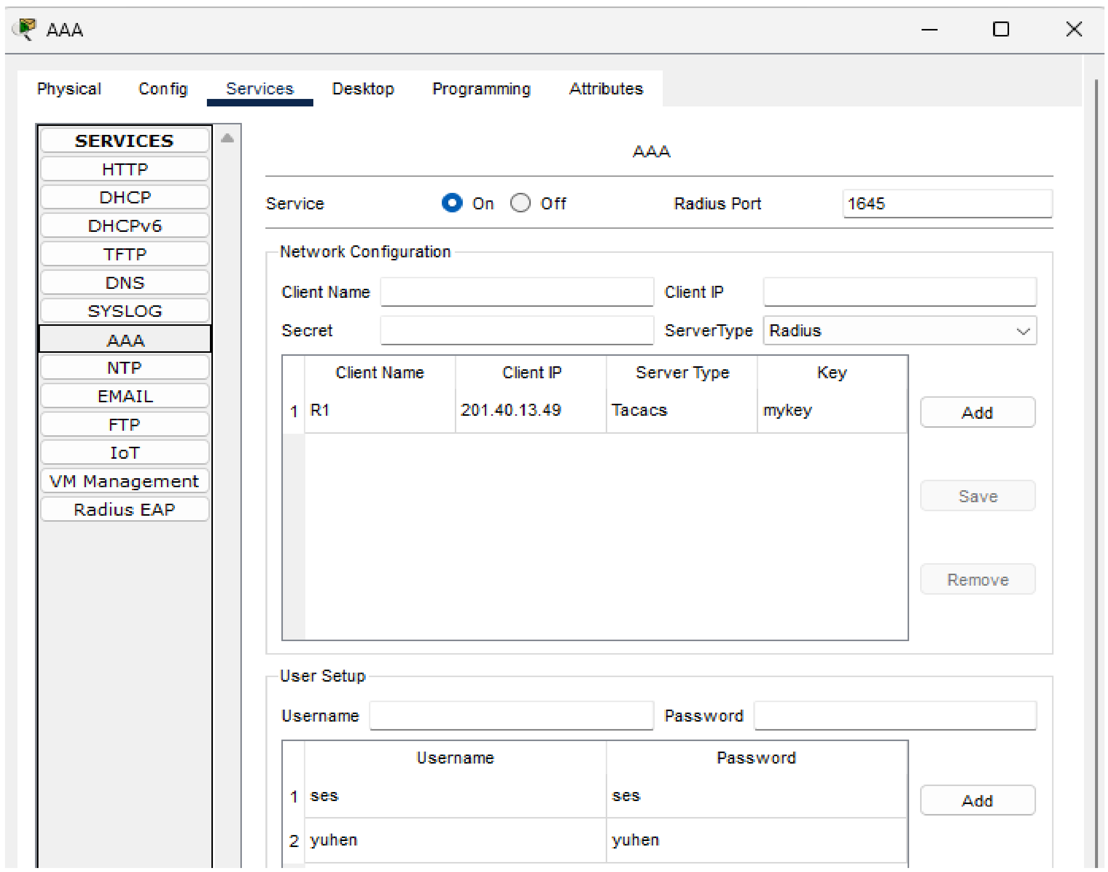
R3

```
R3(config-router)#exit
R3(config)#
R3(config)#ip route 201.40.13.56 255.255.255.252 s2/0
R3(config)#
```

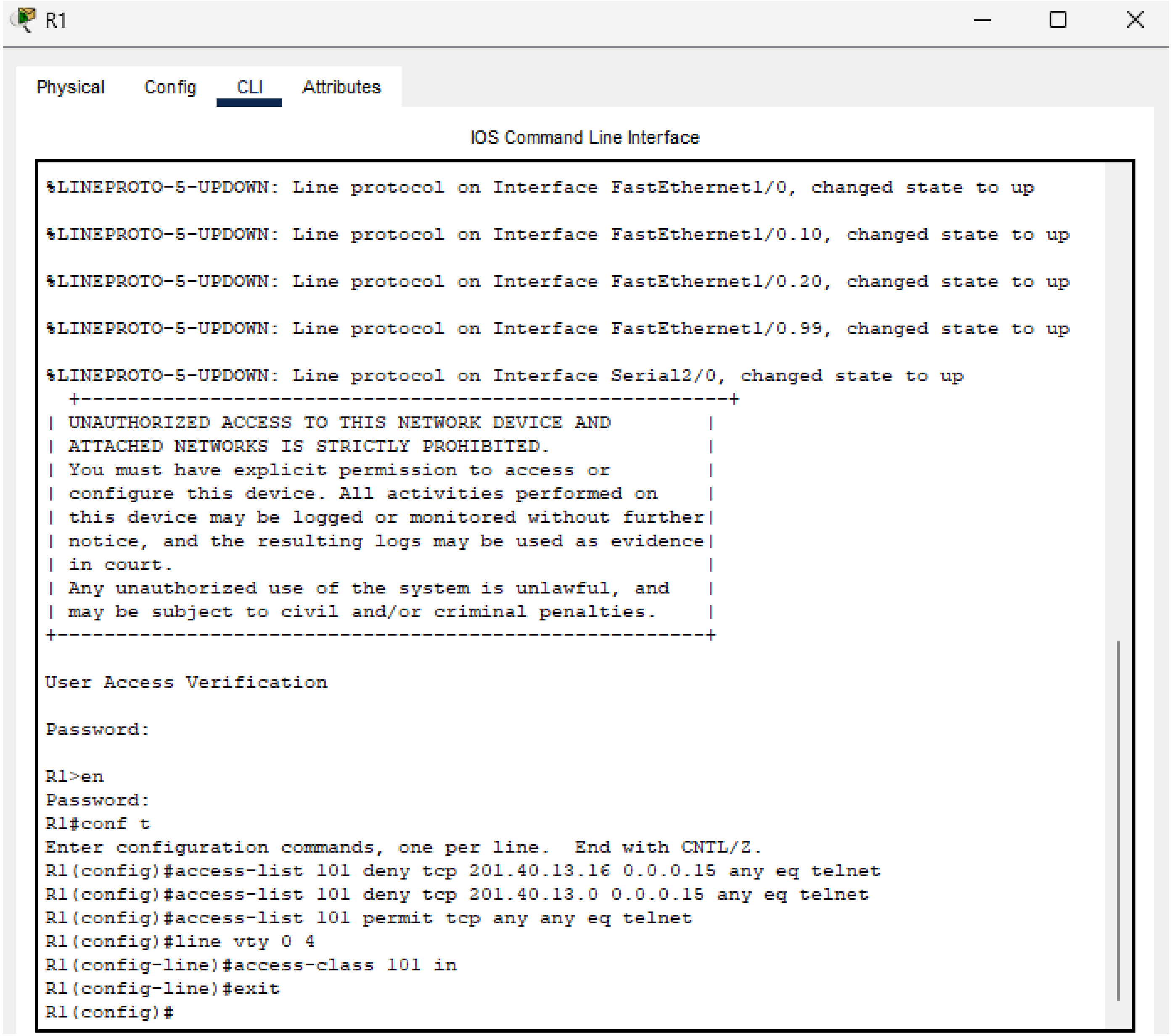
R1 AAA CONFIGURATION

```
User Access Verification

Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#aaa new-model
R1(config)#aaa authentication login myauth group tacacs+
local
R1(config)#tacacs-server host 201.40.13.52 key mykey
R1(config)#line vty 0 4
R1(config-line)#login authentication myauth
R1(config-line)#
```

AAA SERVER



R1 ACCESS-LIST




```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
  +----------------------------------------------------------+
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND         |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.              |
| You must have explicit permission to access or         |
| configure this device. All activities performed on     |
| this device may be logged or monitored without further|
| notice, and the resulting logs may be used as evidence|
| in court.                                              |
| Any unauthorized use of the system is unlawful, and    |
| may be subject to civil and/or criminal penalties.     |
+----------------------------------------------------------+

User Access Verification

Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 101 deny tcp 201.40.13.16 0.0.0.15 any eq telnet
R1(config)#access-list 101 deny tcp 201.40.13.0 0.0.0.15 any eq telnet
R1(config)#access-list 101 permit tcp any any eq telnet
R1(config)#line vty 0 4
R1(config-line)#access-class 101 in
R1(config-line)#exit
R1(config)#
```